dirac16complex as dark matter and dark energy: symbols and their definitions

A dictionary of every symbol used in the analysis of section 24 of the notebook 7+1.nb, which rests on the solutions of the coupled field equations in the three gravity regimes

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
Document generated by tools/make_docs.py on 2026-10-28

Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one; the names of section 24 begin with `dk`), the defining equation and its meaning, grouped by topic. Longer defining equations follow each table as numbered displayed formulas. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$; energies, masses and temperatures in GeV unless stated; cross sections in cm^2, lifetimes in s; $\hbar c = 1.973269804\times10^{-14}$ GeV cm, $\hbar = 6.582119569\times10^{-25}$ GeV s, 1 GeV $= 1.78266192\times10^{-24}$ g. The uniform Kohn–Sham matter of section 8 below uses units $m = 1$.
- Coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates; on the world brane $x = (x_0, x_1, x_2, x_3)$, and $y = (x_4, \ldots, x_7)$.
- A prime is d/dx_0; τ is the time since the singularity; t the cosmic time of the brane; "ln" the natural logarithm.
- Status labels: DERIVED (an exact consequence of the field equations, checked), ESTIMATE (order of magnitude, assumptions stated), OPEN (not decided by the model).

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The

tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\dots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\dots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1,\dots,e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \;\; (i = 1,\dots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. The model

symbol	Mathematica	definition	meaning
ψ, $\bar\psi$	`dirac16complex`	$\bar\psi = \psi^\dagger\Gamma_0$	the 16-component complex field and its Dirac conjugate
Γ_A, Γ^A, Γ^μ	`Γ17[A]`, `Γ17up[A]`	(2.1)	real 16×16 Dirac matrices; frame matrices
Ω_μ	—	(2.1)	spin connection
m	`mDM` (symbolic, section 24.2)		mass of dirac16complex (free)
g_r, Θ^K, η_K	`gg[r]`	(2.2)	the nine self-couplings; Clifford products; their signs
α_k, Λ, κ	`alf[k]`, `lamb`, `kapS`	$\alpha_0 = 2\Lambda$, $\alpha_1 = 1$, $\kappa = 8\pi G_8 = M_8^{-6}$	Lovelock couplings, bulk cosmological constant
L_2, L_3	—	(2.3)	Gauss–Bonnet and cubic Lovelock densities
$U(b)$, U_0, K_U, b_0	`uPot`, `u0B`, `kU`	(2.4)	stabilizing potential of the auxiliary space
$W = \{y = 0\}$, h	—	$h_{\mu\nu} = g_{\mu\nu}(x,0)$	world brane and its induced metric
f_W	`fWb`		brane tension
$\mathcal{L}_{\rm SM}$, Φ, $\tilde\Phi$	—	$\tilde\Phi = i\sigma_2\Phi^*$	Standard Model Lagrangian on the brane, Higgs doublet
y_{is}, G_P^f, λ_P	—	(2.5)	Yukawa (neutrino), vector and Higgs portals
L_i, ψ_s, f	—		lepton doublets; brane-right-handed parts of ψ at the brane; Standard Model fermions

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \quad \Gamma^A = \eta^{AA}\Gamma_A, \quad \Gamma^\mu = e_A{}^\mu\Gamma^A, \tag{2.1}$$
$$\Omega_\mu = \tfrac{1}{4}\eta_{AC}\omega_\mu{}^C{}_B\Gamma^A\Gamma^B, \quad g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu,$$
$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}, \quad K = (A_1 < \cdots < A_r), \tag{2.2}$$

$$\sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \quad \eta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r},$$
$$L_2 = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \tag{2.3}$$
$$L_3 = \tfrac{1}{8}\delta^{\alpha_1\beta_1\alpha_2\beta_2\alpha_3\beta_3}_{\gamma_1\delta_1\gamma_2\delta_2\gamma_3\delta_3} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} R^{\gamma_2\delta_2}{}_{\alpha_2\beta_2} R^{\gamma_3\delta_3}{}_{\alpha_3\beta_3},$$
$$U(b) = U_0 + \tfrac{K_U}{2}\ln^2(b/b_0), \qquad b^8 = \det(-g_{kl}),\ k,l = 4,\dots,7, \tag{2.4}$$
$$S_P = \int_W d^4x\sqrt{|h|}\Big[-\sum_{i,s} y_{is}\bar{L}_i\tilde{\Phi}P_R\psi_s + \text{h.c.} \tag{2.5}$$
$$- G_P^f(\bar{\psi}\Gamma^\mu\psi)(\bar{f}\gamma_\mu f) - \lambda_P\Phi^\dagger\Phi\,\bar{\psi}\psi\Big].$$

3. Cosmology of the three regimes

symbol	Mathematica	definition	meaning
a, b	`a3`, `b4`; `aS`, `bS` (symbols)	(3.1)	scale factors of ordinary and auxiliary space
H_a, H_b	`Ha`, `Hb`	a'/a, b'/b	Hubble rates
β, m_r, ϕ	`betR`, `rIIm`	$\ln(b/b_0)$; $m_r^2 = \kappa K_U/12$; $\sqrt{12}M_{\rm Pl}\beta$	radion, its mass, canonical radion
ε, p_a, p_b	`feStress[W]`	(3.2)	energy density and pressures in ordinary and auxiliary space
$W(a,b)$	`wW`	(3.2)	comoving energy of a source at fixed occupation numbers
ω, k_a, k_b	`kaP`, `kbP`	(3.3)	comoving energy and momenta of one particle
$V_\perp$, G_4, Λ_4, $M_{\rm Pl}$	`rIIVp`, `rIIMPl`	(3.4)	volume of the auxiliary torus; four-dimensional constants
ρ_4, $\rho_{\rm DM}$	—	$\rho_4 = V_\perp\varepsilon$	energy per brane volume; that of the dark matter
$\delta\beta$	`dkDbeta`	(3.5)	radion displacement by the dark matter today
h_*, p, w_a, w_b	`rIhStar`	(3.6)	matter-dominated cubic regime; equation-of-state ratios
P, Q, $\mathcal{C}$	`rIfl`	(3.7)	Einstein power laws with a fluid
A, ϑ, τ_E, H_E	`rIIaF`, `rIIphF`, `rIItEF`	$\beta \approx A\sin(m_r\tau+\vartheta)/(m_r(\tau-\tau_E))$, $H_E = H_a + 2H_b$	radion oscillation after capture
f	`dkF = rIIfrVals[[2]]`		share of the dirac16complex gas in the energy after capture
H_0, Ω_r, Ω_m, Ω_b, Ω_c, Ω_Λ	`rIIH0g`, `rIIOr`, `rIIOm`, `dkOc`, `rIIOL`	(3.8)	Hubble constant and density fractions today
h	`dkh`	$H_0 = 100h$ km/s/Mpc	reduced Hubble constant (0.674)
H_∞	`rIIIH`	$H_0\Omega_\Lambda^{1/2}$	final de Sitter rate
t_0	`dkAge`	13.79 Gyr	age of the universe

$$ds^2 = dx_0^2 - a^2(dx_1^2+dx_2^2+dx_3^2) - b^2(dx_4^2+\cdots+dx_7^2), \tag{3.1}$$
$$\varepsilon = \frac{W}{a^3b^4}, \qquad p_a = -\frac{\partial_a W}{3a^2b^4}, \qquad p_b = -\frac{\partial_b W}{4a^3b^3}, \qquad \varepsilon' + 3H_a(\varepsilon+p_a) + 4H_b(\varepsilon+p_b) = 0, \tag{3.2}$$
$$\omega = \sqrt{m^2 + k_a^2/a^2 + k_b^2/b^2}, \tag{3.3}$$

$$V_\perp = b_0^4 L_c^4, \qquad 8\pi G_4 = \frac{\kappa}{V_\perp} = \frac{1}{M_{\rm Pl}^2}, \qquad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp}, \qquad 3H_a^2 = 8\pi G_4 \rho_4 + \Lambda_4, \tag{3.4}$$

$$\beta'' + 3H_a\beta' + 4\beta'^2 = \kappa\Big(\frac{\varepsilon}{6} - \frac{p_a}{2} + \frac{p_b}{3}\Big) + \frac{\Lambda}{3}, \qquad \delta\beta = \frac{\kappa\varepsilon}{6m_r^2} = \frac{H_0^2\Omega_c}{2m_r^2}, \tag{3.5}$$

$$h_* = -\sqrt{\alpha_2/(54\alpha_3)}, \qquad a \propto \tau^p, \quad p = \frac{1}{1+w_a}, \qquad p_a = w_a\varepsilon, \quad p_b = w_b\varepsilon, \tag{3.6}$$

$$3P(1+w_a) + 4Q(1+w_b) = 2, \qquad 3(P^2 + 4PQ + 2Q^2) = \mathcal{C}, \qquad a \propto \tau^P,\ b \propto \tau^Q, \tag{3.7}$$

$$H_a = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}, \qquad \Omega_m = \Omega_b + \Omega_c, \qquad \Omega_c = \frac{\Omega_c h^2}{h^2}. \tag{3.8}$$

4. Kaluza–Klein modes of the auxiliary torus

symbol	Mathematica	definition	meaning
$\vec{n}$, $\vec{q}$	—	$\vec{q} = 2\pi\vec{n}/L_c$, $\vec{n} \in \mathbb{Z}^4$	Kaluza–Klein momentum
$\ell_\perp$	`dkLp`	$V_\perp^{1/4}$	size of the auxiliary torus
μ_n, M_n	`rIImuK`	(4.1)	Kaluza–Klein mass and mass of a level
$r_4(k)$, $\sigma(k)$	`rIIr4`, `rIIsig`	(4.2)	number of $\vec{n}$ with $\lvert\vec{n}\rvert^2 = k$; divisor sum
$N(\mu)$, dN	—	(4.3)	number and density of levels
χ, χ_n	—	$\psi(x,y) = \sum_n \chi_n(x) e^{i\vec{q}_n\cdot\vec{y}}/\sqrt{V_\perp}$	four-dimensional fields of the levels (four brane Dirac spinors each); $\chi = \chi_0$ the lightest mode
g	—	16	internal states of the lightest mode (four spinors × two spins × particle, antiparticle)

$$\mu_n = \frac{2\pi|\vec{n}|}{\ell_\perp}, \qquad M_n = \sqrt{m^2 + \mu_n^2}, \tag{4.1}$$

$$r_4(k) = 8\sigma(k) - 32\sigma(k/4), \qquad \sigma(k) = \sum_{d|k} d, \qquad \sigma(k/4) = 0 \text{ unless } 4 \mid k, \tag{4.2}$$

$$N(\mu) = \frac{V_\perp \mu^4}{32\pi^2}, \qquad dN = \frac{V_\perp \mu^3 d\mu}{8\pi^2} \qquad \text{(continuum limit)}. \tag{4.3}$$

5. Observational quantities

symbol	Mathematica	definition	meaning
Y	—	n/s	comoving number density (number per entropy)
mY	`dkYm`	(5.1)	required mass per entropy of the dark matter
ρ_c, s_0, g_{*s0}, $T_{\rm CMB}$	`dkRhoc`, `dks0`, `dkGs0`, `rIITcmb`	(5.1)	critical density, entropy density today and its degrees of freedom, CMB temperature
M_D, R, n	`dkMD`; `rRS`	(5.2)	Planck scale and radius in the convention of the Particle Data Group

symbol	Mathematica	definition	meaning
$\sigma_{\rm SI}$	`dkSigSI[c_P, m]`	(6.2)	spin-independent dark matter–nucleon cross section
σ/m	`dkSigM[c_r, m]`	(6.3)	self-interaction cross section per mass
w_0, w_a	—	$w(a) = w_0 + w_a(1-a)$	parameters of evolving dark energy (DESI)
$m_{\rm WDM}$	—		mass of a thermal-relic warm dark-matter particle (Lyman-α bound)
Pauli bound	`dkPauli`	$0.13\,\mathrm{keV}\,(2/g)^{1/4}$	phase-space bound on fermionic dark matter

$$mY = \frac{\Omega_c h^2\,(\rho_c/h^2)}{s_0}, \qquad \frac{\rho_c}{h^2} = 3\Big(100\,\frac{\rm km}{\rm s\,Mpc}\Big)^2 M_{\rm Pl}^2, \qquad s_0 = \frac{2\pi^2}{45} g_{*s0} T_{\rm CMB}^3, \qquad g_{*s0} = 2 + \frac{21}{4}\cdot\frac{4}{11}, \tag{5.1}$$

$$M_{\rm Pl}^2 = R^n M_D^{n+2}, \quad V_\perp = (2\pi R)^n \ (n=4) \quad \Rightarrow \quad M_D = (2\pi)^{2/3} M_8. \tag{5.2}$$

6. The lightest mode: couplings, detection, stability

symbol	Mathematica	definition	meaning
c_P, c_y, c_V, c_r	`cPS`, `cyS`, `cVS`, `crS`	(6.1)	dimensionless couplings in units of M_8 (natural size: 1)
$\lambda_{\rm eff}$, $y_{\rm eff}$, $G_{\rm eff}$, G_r	`dkLam`, `dkYeff`, —, `dkGr`	(6.1)	four-dimensional couplings of a level
v, m_h, G_F, α	`vEW`, `mHb`, `gFermi`	$v = (\sqrt{2}G_F)^{-1/2}$	Higgs vacuum value, Higgs mass, Fermi constant, fine-structure constant
f_N, m_N, μ	—	0.30, 0.9383 GeV, $mm_N/(m+m_N)$	Higgs–nucleon coupling factor, nucleon mass, reduced mass
$G_{\chi N}$	—	$\lambda_{\rm eff} f_N m_N / m_h^2$	contact coupling of χ to nucleons
μ_P, θ	`dkMuP[c_y]`	$\mu_P = y_{\rm eff} v/\sqrt{2}$, $\theta = \mu_P/m$	portal Dirac mass; neutrino mixing angle
$\Gamma(\chi\to\nu\gamma)$, $\Gamma(\chi\to 3\nu)$	`dkTauG`, `dkTau3` (lifetimes)	(6.4)	decay rates through the Yukawa portal
K	`dkK`	$\tau(\chi\to 3\nu)\,c_y^2 m^3$	lifetime constant (s GeV^3)
$m_{\rm max}$	`dkMmax`	$(K/(c_y^2 t_0))^{1/3}$	largest mass that outlives the universe ($c_y = 1$)

$$\lambda_P = \frac{c_P}{M_8^5}, \quad y = \frac{c_y}{M_8^2}, \quad G_P = \frac{c_V}{M_8^6}, \quad g_r = \frac{c_r}{M_8^6}; \tag{6.1}$$

$$\lambda_{\rm eff} = \frac{c_P M_8}{M_{\rm Pl}^2}, \quad y_{\rm eff} = \frac{c_y M_8}{M_{\rm Pl}}, \quad G_{\rm eff} = \frac{c_V}{M_{\rm Pl}^2}, \quad G_r = \frac{c_r}{M_{\rm Pl}^2},$$

$$\sigma_{\rm SI} = \frac{\lambda_{\rm eff}^2 f_N^2 m_N^2 \mu^2}{\pi m_h^4}, \tag{6.2}$$

$$\sigma \approx \frac{G_r^2 m^2}{4\pi}, \tag{6.3}$$

$$\Gamma(\chi\to\nu\gamma) = \frac{9\alpha G_F^2 m^5 \sin^2 2\theta}{1024\pi^4}, \qquad \Gamma(\chi\to 3\nu) = \frac{G_F^2 m^5 \theta^2}{96\pi^3}, \tag{6.4}$$

$$\frac{\Gamma(\chi \to \nu\gamma)}{\Gamma(\chi \to 3\nu)} = \frac{27\alpha}{8\pi}.$$

7. The abundance

symbol	Mathematica	definition	meaning
O	`dkOver[T_RH]`	(7.1)	overabundance of the primordial gas over the dark matter
$T_{\rm RH}$	`trh`		reheating temperature of the brane
g_*, g_{*s}	`rIIgstars`	$\rho_{\rm rad} = \frac{\pi^2}{30} g_* T^4$, $s = \frac{2\pi^2}{45} g_{*s} T^3$	relativistic degrees of freedom
$\Gamma_{\rm brane}$, $\Gamma_{\rm bulk}$, Γ_n	`rIIgam`, `rIIgBulk`	(7.2)	radion decay rates into brane fields and into Kaluza–Klein pairs
$b_{\rm QCD}$, α_s, N_c	—	$11 - \frac{2}{3} n_f$	QCD beta-function coefficient, strong coupling, colours
$B(x, y)$	`Beta`	$\Gamma(x)\Gamma(y)/\Gamma(x+y)$	Euler beta function
T_*	`dkTstar`	$(M_8^6/M_{\rm Pl})^{1/5}$	normalcy temperature
$S(x)$, $x_{\rm RH}$	`dkS[x]`	(7.3)	fraction of the freeze-in yield made below $T_{\rm RH}$; $x_{\rm RH} = m_h/T_{\rm RH}$
K_ν	`BesselK[ν, ·]`		modified Bessel function of the second kind
$M_{\rm Pl}^{(\rm unreduced)}$	1.22091*^19	$\sqrt{8\pi} M_{\rm Pl}$	unreduced Planck mass (freeze-in formula)
N	`nch`		number of decay channels (lepton flavours, brane spinors)
m_*	`dkMstar[N]`	(7.4)	largest mass for which freeze-in and stability are compatible
$Y_{\rm tower}$, $\rho_{\rm tower}/s$	`dkTowN`, `dkTowE`	(7.5)	Kaluza–Klein tower made through the Yukawa portal
I_{hp}, I_{hp}^E	`dkIhp`, `dkIhpE`	(7.6)	phase-space integrals of the Higgs-portal pairs
$\lambda(1, x_1, x_2)$	—	(7.6)	Källén function
T_V	`dkTV`	$(M_8^{12}/M_{\rm Pl})^{1/11}$	temperature above which pairs of gravitational strength are made efficiently

$$O = \frac{f}{1-f}\,\frac{T_{\rm RH}}{\frac{4}{3} m Y}, \qquad \frac{\rho_{\rm DM}}{\rho_{\rm rad}} = \frac{4}{3}\,\frac{mY}{T}, \tag{7.1}$$

$$\Gamma(\phi \to f\bar f) = \frac{N_c m_f^2 m_r}{24\pi M_{\rm Pl}^2}\Big(1 - \frac{4m_f^2}{m_r^2}\Big)^{3/2}, \quad \Gamma(\phi \to gg) = \frac{b_{\rm QCD}^2 \alpha_s^2 m_r^3}{96\pi^3 M_{\rm Pl}^2}, \quad \Gamma_n = \frac{3\mu_n^2 m_r}{8\pi M_{\rm Pl}^2}\Big(1 - \frac{4\mu_n^2}{m_r^2}\Big)^{3/2}, \tag{7.2}$$

$$S(x) = \frac{2}{3\pi}\int_x^\infty t^3 K_1(t)\,dt, \qquad Y = \frac{135\, g_B}{8\pi^3 \cdot 1.66\, g_*^{3/2}}\,\frac{M_{\rm Pl}^{(\rm unreduced)}\Gamma}{m_B^2}\,S(x_{\rm RH}), \tag{7.3}$$

$$m_* = \Big(\frac{Y(c_y = 1)\,K}{mY\,t_0}\Big)^{1/2}, \tag{7.4}$$

$$\sum_n \Big(1 - \frac{\mu_n^2}{m_h^2}\Big)^2 = \frac{V_\perp m_h^4}{192\pi^2}, \qquad \sum_n \mu_n \Big(1 - \frac{\mu_n^2}{m_h^2}\Big)^2 = \frac{V_\perp m_h^5}{315\pi^2}, \tag{7.5}$$

$$I_{hp} = \iint_{u_1+u_2<1} u_1^3 u_2^3 \big(1-(u_1+u_2)^2\big)^{3/2} \big(1-(u_1-u_2)^2\big)^{1/2} du_1 du_2, \tag{7.6}$$

$\lambda(1,x_1,x_2) = (1-x_1-x_2)^2 - 4x_1x_2.$

8. Dark energy

symbol	Mathematica	definition	meaning
ρ, ρ_S, k_F	`rho7[kF]`, `c7 iS7`	(8.1)	density, scalar density, Fermi momentum of uniform matter (7 space dimensions)
m^*, g_S, g_V	`gap7`, `gS0`, `gV0`	(8.1)	effective mass; scalar and vector couplings (-10^5, 10^4)
ε, ε_F, μ	`eps7`, `fermi7`	(8.1)	energy density, Fermi energy, chemical potential
p, w	`dkPres[kF]`, `dkWtab`	(8.2)	pressure and equation of state
k_{F0}, ρ_0	`kF0`	ε/ρ minimal	saturation
ρ_Λ	`dkRhoL`	$3H_0^2 M_{\rm Pl}^2 \Omega_\Lambda$	dark-energy density today
$\rho_{\rm Cas}$	`dkCas[mℓ]`	(8.3)	Casimir (vacuum) energy per brane volume on the torus
Z	`dkZ48`	(8.3)	Epstein zeta value $\sum_{\vec n\neq 0} \lvert\vec n\rvert^{-8}$
$\zeta(s)$	`Zeta`	$\sum_{k\geq 1} k^{-s}$	Riemann zeta function

$$\rho = \frac{k_F^7}{105\pi^4}, \quad c_7 = \frac{1}{15\pi^4}, \quad \rho_S = c_7 \int_0^{k_F} \frac{k^6 m^* dk}{\sqrt{k^2+m^{*2}}}, \quad m^* = m + g_S\rho_S, \tag{8.1}$$

$$\varepsilon = c_7 \int_0^{k_F} k^6 \sqrt{k^2+m^{*2}}\, dk - \frac{(m^*-m)^2}{2g_S} + \frac{g_V}{2}\rho^2, \qquad \varepsilon_F = g_V\rho + \sqrt{k_F^2+m^{*2}},$$

$$p = \rho^2 \frac{d}{d\rho}\Big(\frac{\varepsilon}{\rho}\Big) = \mu\rho - \varepsilon, \qquad \mu = \frac{d\varepsilon}{d\rho} = \varepsilon_F, \qquad w = \frac{p}{\varepsilon}, \tag{8.2}$$

$$\rho^{(1)}_{\rm Cas} = -\frac{2}{\ell_\perp^4} \sum_{\vec n\neq 0} \Big(\frac{m\ell_\perp}{2\pi|\vec n|}\Big)^4 K_4(m\ell_\perp|\vec n|) \xrightarrow{m\to 0} -\frac{\Gamma(4)Z}{\pi^4\ell_\perp^4}, \tag{8.3}$$

$$Z = \sum_{k\geq 1} \frac{r_4(k)}{k^4} = 8(1-4^{-3})\zeta(4)\zeta(3).$$

$\rho^{(1)}_{\rm Cas}$ is per real bosonic degree of freedom with periodic boundary conditions; a fermion contributes with the opposite sign; dirac16complex has 16 real degrees of freedom on shell ($\rho_{\rm Cas} = -16\rho^{(1)}_{\rm Cas}$), the eight-dimensional graviton 20.

9. Symbols that look alike

- W: the world brane $\{y=0\}$, and $W(a,b)$, the comoving energy of a source (always written with its arguments).
- h: the induced metric $h_{\mu\nu}$ of the brane, the reduced Hubble constant $h = 0.674$, and the Higgs boson h (mass m_h).
- μ: the reduced mass $mm_N/(m+m_N)$ (section 6), the chemical potential (section 8), and with an index the Kaluza–Klein mass μ_n and the portal mass μ_P.
- λ: the Higgs-portal coupling λ_P and $\lambda_{\rm eff}$, the Källén function $\lambda(1,x_1,x_2)$; the Higgs self-coupling is not used here.

- N: the number of decay channels; $N(\mu)$ the number of Kaluza–Klein levels lighter than μ; N_c the number of colours.
- S: the Boltzmann factor $S(x)$ (section 7), and S_P, the portal term of the action, (2.5).
- K: the lifetime constant K (section 6); K_ν the Bessel function; K_U the stiffness of $U(b)$; K as a multi-index in Θ^K.
- σ: the cross sections $\sigma_{\rm SI}$, σ; the divisor sum $\sigma(k)$; σ_r the sign exponent in Θ^K; σ_2 the Pauli matrix.
- g: the number of internal states (Pauli bound), g_*, g_{*s}, g_B (degrees of freedom), g_r (self-couplings), g_S, g_V (Kohn–Sham couplings), g_n (radion couplings of the levels), $g_{\mu\nu}$ (metric).
- Γ: decay rates; Γ_A the Dirac matrices; $\Gamma(4) = 6$ the Gamma function.

10. Mathematica names of section 24

name	**meaning**
`dkOch2`, `dkh`, `dkOc`, `dkHbarC`	$\Omega_c h^2 = 0.120$, $h = 0.674$, Ω_c, $\hbar c$ in GeV cm
`dkRhoc`, `dkGs0`, `dks0`, `dkYm`	ρ_c/h^2, g_{*s0}, s_0, mY
`dkMD`, `dkPauli`	M_D; Pauli bound for $g = 16$ (keV)
`dkSt`, `mDM`, `kaP`, `kbP`	stress tensor of a massive particle; its mass and comoving momenta (symbols)
`dkDbeta`	$\delta\beta$
`dkLam`, `dkYeff`, `dkGr`, `dkGeV2g`	$\lambda_{\rm eff}$, $y_{\rm eff}$, G_r per unit coupling; grams per GeV
`dkSigSI`, `dkSigM`, `dkCrMax`	$\sigma_{\rm SI}$, σ/m, largest c_r allowed by the Bullet Cluster
`dkMuP`, `dkTauG`, `dkTau3`, `dkAge`, `dkMmax`	μ_P, lifetimes, t_0 in s, $m_{\max}$
`dkF`, `dkOver`, `dkFracBulk`	f, $O(T_{\rm RH})$, $\Gamma_{\rm bulk}/(\Gamma_{\rm bulk} + \Gamma_{\rm brane})$
`dkTstar`, `dkS`, `dkSstar`, `dkC`	T_*, $S(x)$, $S(m_h/T_*)$, the freeze-in prefactor
`dkYfi`, `dkYhp`, `dkK`, `dkMstar`, `dkCyNeed`, `dkCyMax`	zero-mode yields (Yukawa, Higgs portal), K, m_*, c_y needed and allowed
`dkTowN`, `dkTowE`, `dkLat`, `dkLatE`, `dkCyTow`	tower sums (continuum), lattice sums ($R = 100$), c_y bound
`dkIhp`, `dkIhpE`, `dkPairs`, `dkRhp`, `dkCpNeed`	I_{hp}, I^E_{hp}, $(V_\perp m_h^4/(8\pi^2))^2$, mass per entropy of the pairs, c_P needed
`dkTV`	T_V
`dkPE`, `dkPres`, `dkWtab`, `dkWmin`, `dkKmin`	(p, ε), p, table of (k_F, p, w), $w_{\min}$ and its k_F
`dkZ48`, `dkLp`, `dkRhoL`, `dkC0`, `dkCas`, `dkXL`	Z, $\ell_\perp$, ρ_Λ, $\Gamma(4)Z/\pi^4$, $\rho_{\rm Cas}$, $m\ell_\perp$ at $\rho_{\rm Cas} = \rho_\Lambda$